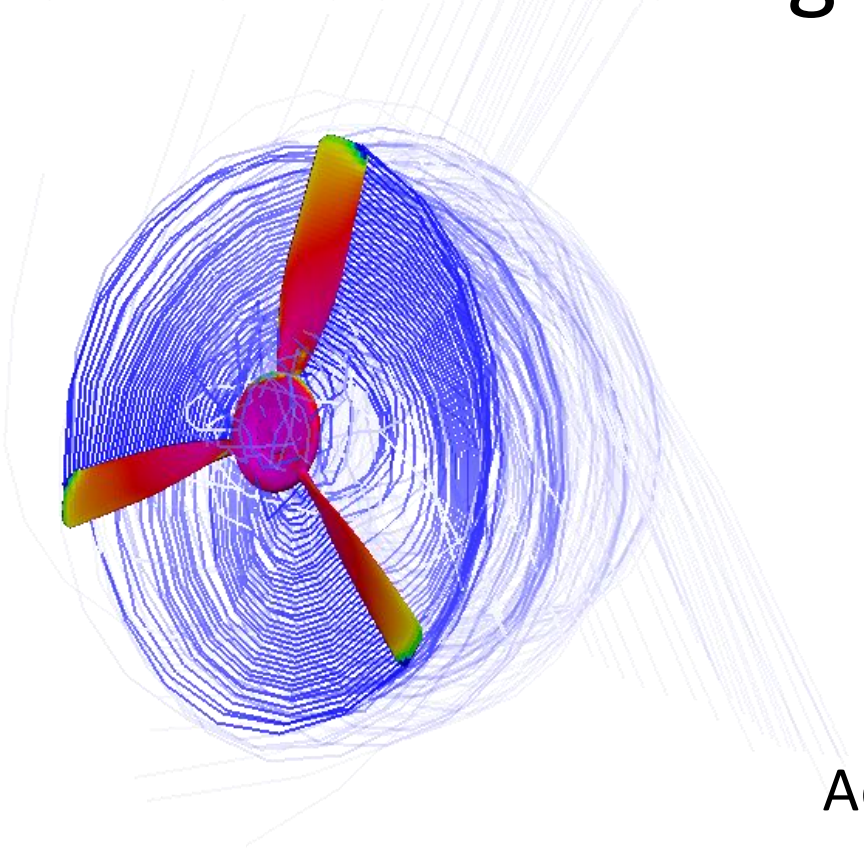
LIÈGE université
Sciences Appliquées

Introduction to Propeller design and performance modeling



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Academic year 2025-2026

Propeller project – Context

Distributed electric propulsion

- Multiple small propellers
- Strong wing-propeller interactions
- Increased takeoff lift

Traditional propulsion

- One or two large propellers
- Limited aero-propulsive interactions
- Lift mainly aerodynamic

DEP reduces the required takeoff power

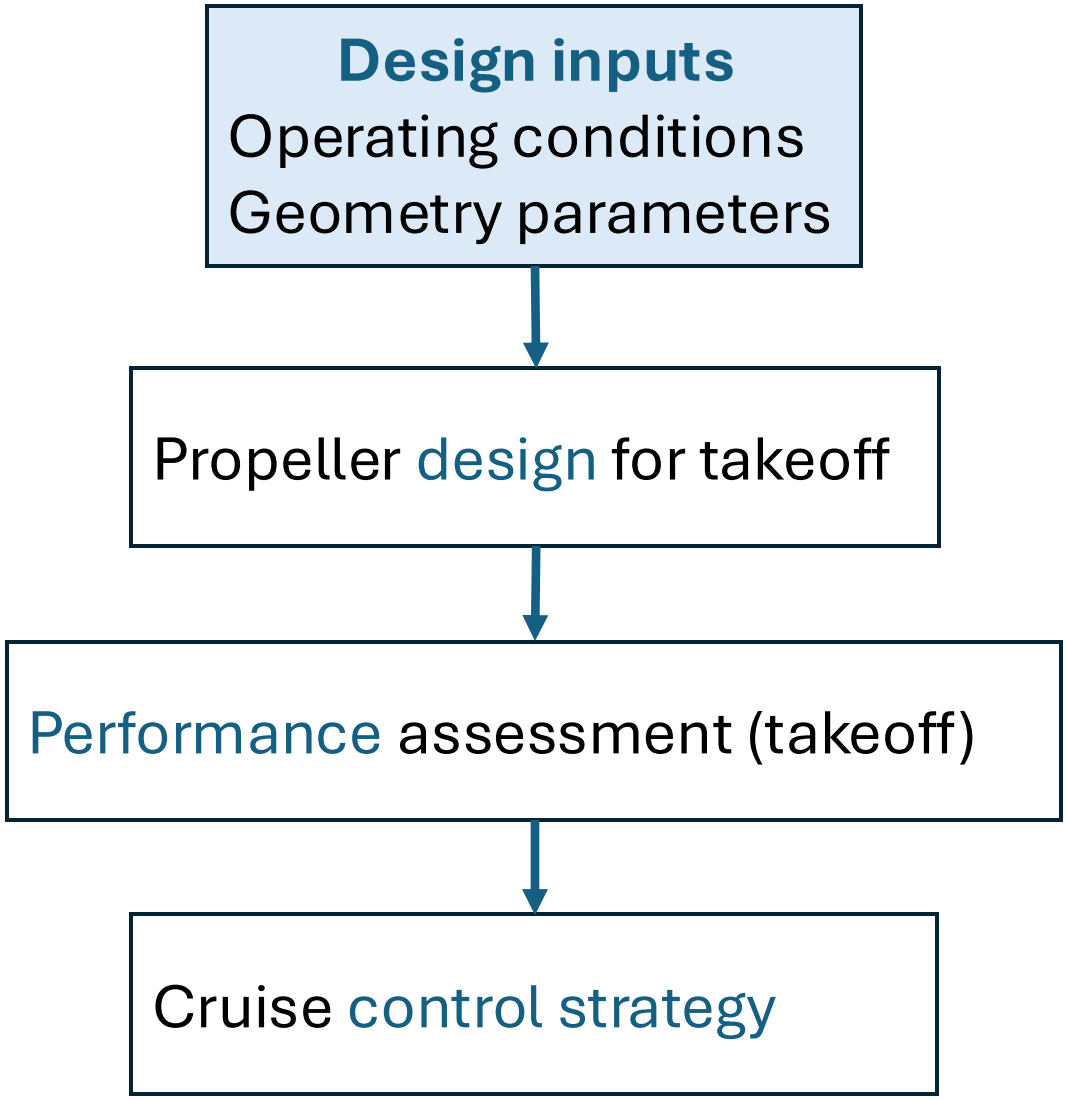
Propeller project – Big picture

Design phase

1. Implement Adkins & Liebeck
2. Design the propeller
3. Obtain the twist and chord distributions
4. Plot and discuss your results

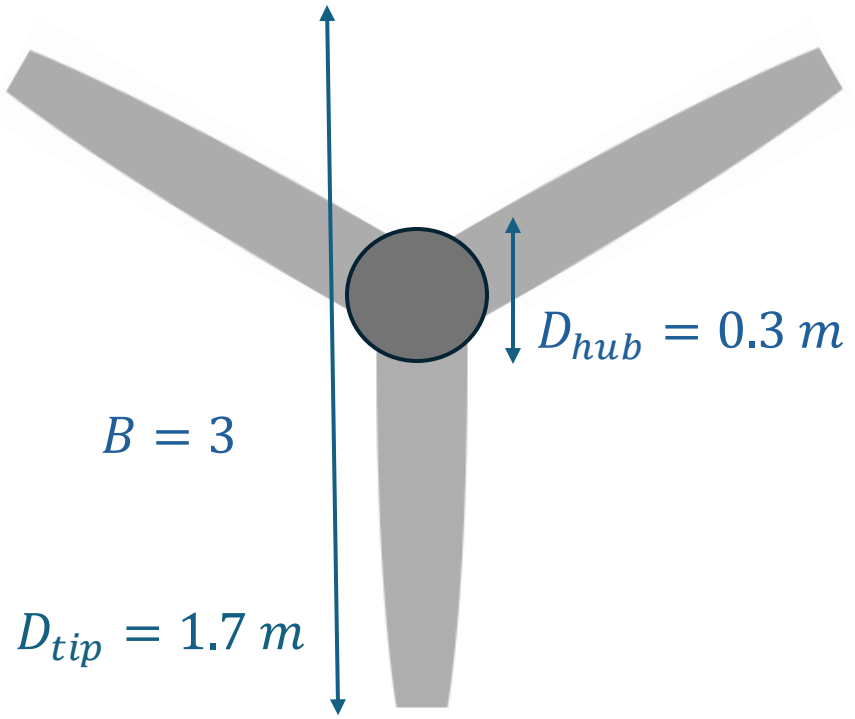
Performance assessment

1. Implement the BEMT
2. Evaluate takeoff performance
3. Interpret using velocity triangles
4. Plot the performance coefficient maps
5. Use control strategies for cruise



Propeller project – Geometry & operation

Geometry



Airfoil: Clark-Y

Operating conditions

Takeoff:

- Propeller speed: $\mathrm{n} = 2100\ RPM$
- Free-stream velocity: $V_\infty = 45\ m/s$
- Sea-level conditions

Cruise:

- Propeller speed: $\mathrm{n} = 2100\ RPM$
- Free-stream velocity: $V_\infty = 90\ m/s$
- Sea-level conditions

Propeller project – Deliverables & grading

Project organization

- Groups of 2 students
- Deadline: 16 May 2026
- Documents on eCampus
  - Project statement
  - Airfoil polars
  - Reference paper

Submission

- Code archive (.zip) + README
- PDF report (max 20 pages)

Evaluation

- 30% of course grade

References

C. N. Adkins and R. H. Liebeck. “Design of optimum propellers”. In: Journal of Propulsion and Power 10.5 (1994),pp. 676–682.

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